

Advanced Algorithms [MC4203]

Fall Semester, 2023

Department	Mathematics and Computer Science	Credits	3	Instructor	Hyunwoo Jung	Class Room	Creation Building(CB) 702
Subject	Advanced Algorithms	Class Hours/Wk	3	E-mail			
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Office Phone	051-606-2350			Office Hour	6 th period on Mon		
Office	Changjo 703						

Instructors would be available at any time after a prior appointment.

1. Course Description

Course Objectives	
This is an advanced algorithm course. Students learn practical algorithm in the state of the art. Also students learn how to analyze time and space complexity mathematically for those algorithms which have low complexity. Student will understand how to design algorithms for real life problems and how to verify their efficiency thoroughly using mathematical analysis. We introduce numerous algorithms in various fields. We make students have professional knowledge and utmost problem solving abilities.	
Contents	
1. Probabilistic Method i. Random Variable and Expectation ii. Markov's Inequality iii. Chebyshev's Inequality iv. Derandomization 2. Entropy, Randomness, and Information i. Entropy Function ii. Shannon's Theorem 3. Primal Dual Algorithms i. Linear Programming ii. Duality Theorem iii. Primal Dual Algorithms iv. Primal Dual Approximation Algorithms	4. Advanced Algorithms i. Simplex Method ii. Lovasz Local Lemma iii. Shor's Factorization Algorithm iv. Online Algorithm(K-servers on a line) v. VCG Mechanism vi. Buchberger's Algorithm vii. Local Search Algorithm for Facility Location

2. Prerequisites

CS1, CS2, CS3, Algorithm, Data Structure, etc.

3. Course Info

No main textbook

Main materials can be downloaded from the course link or distributed during the lecture:

Course Link

Google Classroom

4. Grading Policy

Individual Presentation	Homework Assignments	Individual Project	Attendance & Attitude
20%	40%	30%	10%

Individual Presentation

A. We encourage students to present solution for homework problems and in-class problems.

Individual Presentation

A. Students present one of the advanced topics suggested in the lecture.

Attendance & Attitude

A. Bad attitudes such as sleeping, chatting, etc. result in deduction.

5. Weekly Schedule

Weeks	Topics
Week1	Random Variables, Expectation
Week2	Markov's Inequality Chebyshev's Inequality
Week3	Derandomization
Week4	Entropy, Shannon's Coding Theorem
Week5,6,7	Linear Programming, Primal-Dual Algorithms, Primal-Dual Approximation Algorithms
Week8	Midterm Exam
Week9	Simplex Method
Week10	Lovasz Local Lemma
Week11	Shor's Factorization Algorithm
Week12	Online Algorithm(k-servers on a line)
Week13	VCG Mechanism
Week14	Buchberger's Algorithm
Week15	Local Search Algorithm for Facility Location
Week16	Final Exam Period

*** The schedule will change as it may chance.

6. Homework submission due date

There is a deadline for each assignment.

7. Policy for cheating

No cheating activity is allowed. Students who cheat, plagiarize, copy, fake, etc. will fail the course.

8. Proportion of A or B Grades

Basically, the proportion of students who get A or B grade is 70%. But if all students get high achievements, this ratio would be raised.